CS 577 Deep Learning
Homework 3: Attention and Transformer Forward

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Part 0: Setup and Axes

All experiments were executed on a CPU with a fixed random seed of `577`. The notebook environment used the following versions:

- Python: `3.11.18`
- NumPy: `2.4.6`
- PyTorch: `2.15.0`
- Matplotlib: `3.11.2`
- Device: `CPU`

All 11 public tests passed successfully. These tests covered stable masked softmax, causal masking, scaled dot-product attention, head splitting and merging, multi-head self-attention and cross-attention, positional encodings, and Transformer block behavior.

The final test summary was:

```
Ran 11 tests in 0.046s

OK

Public-test return code: 0
```

No failures or errors were reported.

0.2 Axis and mask contract

The attention scores S, attention weights A, and output Y have the following shapes:

$$S : (B, H, T_q, T_k), \qquad A : (B, H, T_q, T_k), \qquad Y : (B, H, T_q, d_v).$$

Here, B is the batch size, H is the number of heads, T_q is the number of query tokens, T_k is the number of key tokens, d_k is the query/key feature size, and d_v is the value feature size.

We apply softmax along the last axis (T_k) because each query compares itself with all keys. Softmax gives a probability to each allowed key, and these probabilities add up to 1. If all keys are blocked, the probabilities are all zero.

The key mask first has shape (B, T_k). We reshape it to $(B, 1, 1, T_k)$ because each batch may have a different mask, but the same mask is used for all heads and queries within that batch.

NumPy broadcasting works by matching dimensions from right to left. A dimension of size 1 can expand to match the corresponding dimension in another array.

This allows the mask with shape $(B, 1, 1, T_k)$ to work with attention scores of shape (B, H, T_q, T_k).

Part 1: Hand Calculation

The given query, key, and value matrices are:

$$Q = K = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}, \qquad V = \begin{bmatrix} 1 & 0 \\ 0 & 2 \\ 3 & 1 \end{bmatrix}.$$

All three matrices have shape $(3, 2)$, and $d_k = d_v = 2$.

First, we calculate the dot products between queries and keys:

$$QK^\top = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 2 \end{bmatrix}.$$

The shape of $QK^\top$ is $(3, 3)$.

Next, we divide each score by $\sqrt{d_k} = \sqrt{2}$:

$$S = \frac{QK^\top}{\sqrt{2}} = \begin{bmatrix} 0.7071 & 0 & 0.7071 \\ 0 & 0.7071 & 0.7071 \\ 0.7071 & 0.7071 & 1.4142 \end{bmatrix}.$$

The scaled score matrix S has shape $(3, 3)$.

We apply softmax along each row to calculate the attention weights:

$$A = \text{softmax}(S) = \begin{bmatrix} 0.4011 & 0.1978 & 0.4011 \\ 0.1978 & 0.4011 & 0.4011 \\ 0.2483 & 0.2483 & 0.5035 \end{bmatrix}.$$

The attention weight matrix A has shape $(3, 3)$. Each row sums to one.

Softmax calculation for cat. The scaled scores for the cat row are:

$$s_{\text{cat}} = \left[0, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right].$$

The softmax formula is:

$$a_j = \frac{e^{s_j}}{\sum_k e^{s_k}}.$$

The common denominator is:

$$e^0 + e^{1/\sqrt{2}} + e^{1/\sqrt{2}}.$$

Using this denominator, the attention weights are:

$$a_{\text{The}} = \frac{e^0}{e^0 + 2e^{1/\sqrt{2}}} \approx 0.1978,$$

$$a_{\text{cat}} = a_{\text{sleeps}} = \frac{e^{1/\sqrt{2}}}{e^0 + 2e^{1/\sqrt{2}}} \approx 0.4011.$$

The attention weights for cat are:

$$a_{\text{cat,row}} = [0.1978,\ 0.4011,\ 0.4011].$$

Attention output. The attention output is calculated by multiplying the attention weights by the value matrix:

$$Y = AV = \begin{bmatrix} 1.6044 & 0.7967 \\ 1.4011 & 1.2033 \\ 1.7587 & 1.0000 \end{bmatrix}.$$

The output matrix Y has shape $(3, 2)$.

Weighted sum for cat: The value vectors are:

$$V_{\text{The}} = [1, 0], \qquad V_{\text{cat}} = [0, 2], \qquad V_{\text{sleeps}} = [3, 1].$$

We multiply each value vector by its attention weight and add the results:

$$Y_{\text{cat}} = 0.1978[1, 0] + 0.4011[0, 2] + 0.4011[3, 1].$$

Calculating each output component separately gives:

$$Y_{\text{cat}} = [0.1978(1) + 0.4011(0) + 0.4011(3),$$

$$0.1978(0) + 0.4011(2) + 0.4011(1)]\,.$$

The final output for cat is:

$$\boxed{Y_{\text{cat}} = [1.4011,\ 1.2033]}$$

1.2 Causal attention

In causal attention, each query can attend only to the current token and previous tokens. Future tokens are blocked.

In our example:

- The can attend only to The.
- cat can attend to The and cat.
- sleeps can attend to all three tokens.

We use the causal mask M, where 1 means allowed and 0 means blocked:

$$M = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}.$$

The mask has shape $(3, 3)$.

After applying the mask and softmax, the attention weights are:

$$A_{\text{causal}} = \begin{bmatrix} 1.0000 & 0 & 0 \\ 0.3302 & 0.6698 & 0 \\ 0.2483 & 0.2483 & 0.5035 \end{bmatrix}.$$

The weight matrix has shape $(3, 3)$. Blocked keys have attention weights of exactly zero.

The new attention output is:

$$Y_{\text{causal}} = A_{\text{causal}} V = \begin{bmatrix} 1.0000 & 0 \\ 0.3302 & 1.3395 \\ 1.7587 & 1.0000 \end{bmatrix}.$$

The output matrix has shape $(3, 2)$.

Weighted sum for cat. The cat query can attend only to The and cat. The sleeps token is blocked.

The attention weights for cat are:

$$a_{\text{cat}} = [0.3302,\ 0.6698,\ 0].$$

Using the value vectors, we calculate:

$$Y_{\text{cat}} = 0.3302[1, 0] + 0.6698[0, 2] + 0[3, 1].$$

Calculating the two output components separately:

$$\begin{aligned} Y_{\text{cat}} &= \big[0.3302(1) + 0.6698(0) + 0(3), \\ &\qquad 0.3302(0) + 0.6698(2) + 0(1)\big] \\ &\approx [0.3302,\ 1.3396]. \end{aligned}$$

Why the last row does not change: The last token, sleeps, can attend to all three tokens because there are no future tokens after it. Therefore, its attention weights and output remain the same as in unmasked attention.

Why each output has two values: Each value vector has two features $d_v = 2$. Attention calculates a weighted sum of these vectors. The causal mask changes which tokens are used, but it does not change the number of output features. That's why each output vector still has two values.

1.3 Scaling and stability

The dot product between a query and a key is:

$$s = q^\top k = \sum_{r=0}^{d_k-1} q_r k_r.$$

We assume that all coordinates are mutually independent, have zero mean, and unit variance:

$$E[q_r] = E[k_r] = 0, \qquad \mathrm{Var}(q_r) = \mathrm{Var}(k_r) = 1.$$

Using the variance formula and the zero mean:

$$\begin{aligned} \mathrm{Var}(q_r) = E[q_r^2] - (E[q_r])^2 = 1, \\ E[q_r^2] = 1. \end{aligned}$$

Similarly, $E[k_r^2] = 1$.

For each product, independence gives:

$$E[q_r k_r] = E[q_r]E[k_r] = 0.$$

$$E[q_r^2 k_r^2] = E[q_r^2]E[k_r^2] = 1.$$

Therefore,

$$\begin{aligned} \mathrm{Var}(q_r k_r) &= E[(q_r k_r)^2] - (E[q_r k_r])^2 \\ &= 1 - 0^2 \\ &= 1. \end{aligned}$$

Since all coordinates are mutually independent, each product uses a different independent pair. Therefore, the variances of the products add:

$$\begin{aligned}\mathrm{Var}(s) &= \mathrm{Var}\left(\sum_{r=0}^{d_k-1} q_r k_r\right) \\ &= \sum_{r=0}^{d_k-1} \mathrm{Var}(q_r k_r) \\ &= \sum_{r=0}^{d_k-1} 1 \\ &= \boxed{d_k}.\end{aligned}$$

After scaling by $\sqrt{d_k}$:

$$s_{\text{scaled}} = \frac{s}{\sqrt{d_k}}.$$

Using $\mathrm{Var}(cX) = c^2 \mathrm{Var}(X)$:

$$\begin{aligned}\mathrm{Var}(s_{\text{scaled}}) &= \mathrm{Var}\left(\frac{s}{\sqrt{d_k}}\right) \\ &= \frac{1}{d_k}\mathrm{Var}(s) \\ &= \frac{d_k}{d_k} \\ &= \boxed{1}.\end{aligned}$$

Why scaling is important: Without scaling, the standard deviation of the attention scores is $\sqrt{d_k}$. As d_k grows, the scores can become more spread out.

For two softmax weights, their ratio is:

$$\frac{a_i}{a_j} = e^{s_i - s_j}.$$

This means the ratio grows exponentially when the difference between scores increases. As a result, the largest score can receive almost all the attention weight, while the others receive values close to zero.

Dividing by $\sqrt{d_k}$ keeps the score variance controlled and reduces this problem.

Numerical stability: For numerical stability, we subtract the maximum score from each row before applying softmax:

$$m = \max_j s_j.$$

The softmax formula becomes:

$$\begin{aligned}\text{softmax}(s_i) &= \frac{e^{s_i}}{\sum_j e^{s_j}} \\ &= \frac{e^{s_i - m} e^{m}}{\sum_j e^{s_j - m} e^{m}} \\ &= \frac{e^{s_i - m}}{\sum_j e^{s_j - m}}.\end{aligned}$$

The common factor e^m cancels, so subtracting the maximum does not change the softmax result.

After subtraction, the largest shifted score is zero, and its exponential is:

$$e^0 = 1.$$

All other shifted scores are less than or equal to zero, so their exponentials are at most one.

For a row with at least one allowed key, the denominator contains at least one term equal to one. Therefore, it is at least one, which avoids division by zero and denominator underflow.

For example, consider the scores:

$$[1000,\ 999,\ -1000].$$

Computing e^{1000} directly can cause overflow. After subtracting the maximum, we obtain:

$$[0,\ -1,\ -2000].$$

The exponentials are now:

$$[1,\ e^{-1},\ e^{-2000}].$$

This makes the calculation numerically stable without changing the softmax probabilities.

A fully masked row is handled separately by returning zeros, as required by the implementation.

Part 2: Stable Masked Attention

2.1 Numerical stability and fully blocked rows

I implemented masked softmax using NumPy. In my code, True means allowed and False means blocked.

The mask is applied before normalization. Blocked positions receive attention weights of exactly zero.

To avoid overflow, I subtract the largest allowed score in each row before calculating the exponential.

$$A_{ij} = \begin{cases} \dfrac{e^{S_{ij} - m_i}}{\sum_{r \text{ allowed}} e^{S_{ir} - m_i}}, & \text{if } j \text{ is allowed,} \\ 0, & \text{if } j \text{ is blocked.} \end{cases}$$

Here, m_i is the largest allowed score in row i.

For a fully blocked row, the maximum would be $-\infty$. Subtracting it from itself would produce NaN. To prevent this, my code replaces a non-finite row maximum with zero before calculating the exponential.

It also uses a safe denominator and returns zeros for the fully blocked row.

I tested extreme scores, including $[1000, 999, -1000]$, and a fully blocked row.

The results were:

$$A = \begin{bmatrix} 0.7311 & 0.2689 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

The row sums were:

$$[1.0,\ 0.0].$$

All results were finite, with no warnings or NaN values.

My code raises ValueError if a mask is not Boolean or cannot broadcast to the score shape.

2.2 Hand-case agreement

I implemented scaled dot-product attention using NumPy and the following formulas:

$$S = \frac{QK^\top}{\sqrt{d_k}}, \qquad A = \text{masked_softmax}(S), \qquad Y = AV.$$

The input shapes are:

$$Q : (B, H, T_q, d_k),$$
$$K : (B, H, T_k, d_k),$$
$$V : (B, H, T_k, d_v).$$

The function returns (Y, cache), where:

$$Y : (B, H, T_q, d_v).$$

The cache contains the scaled scores before masking and the attention weights. Both have shape:

$$(B, H, T_q, T_k).$$

The implementation supports rectangular attention ($T_q \neq T_k$) and different feature sizes ($d_k \neq d_v$).

It raises `ValueError` when input dimensions are not compatible.

Hand-case results: I tested my implementation using the provided `hand_case()` with three tokens: The, cat, and sleeps.

The inputs have shape $(1, 1, 3, 2)$, which means one batch, one attention head, three tokens, and two features.

For unmasked attention, I obtained:

$$Y_{\text{full}} = \begin{bmatrix} 1.6044 & 0.7967 \\ 1.4011 & 1.2033 \\ 1.7587 & 1.0000 \end{bmatrix}.$$

For causal attention, I obtained:

$$Y_{\text{causal}} = \begin{bmatrix} 1.0000 & 0 \\ 0.3302 & 1.3395 \\ 1.7587 & 1.0000 \end{bmatrix}.$$

Both outputs have shape $(1, 1, 3, 2)$. The matrices above omit the batch and head axes for easier reading.

Numerical agreement: I compared the NumPy results with the values written to four decimal places in Part 1.

The maximum absolute differences, based on the displayed NumPy values, were approximately:

$$\text{Unmasked difference} \approx 4.8 \times 10^{-5},$$
$$\text{Causal difference} \approx 3.8 \times 10^{-5}.$$

Both differences are below 5×10^{-5}, which is the expected rounding difference for four decimal places.

The calculations in NumPy use full precision.

2.3 Padding and causal masks

I tested two sequences with a maximum length of four tokens. The first sequence has two real tokens, and the second has four real tokens.

The padding mask has shape $(B, 1, 1, T)$. The causal mask has shape $(1, 1, T, T)$.

I combined them using Boolean AND:

$$M_{\text{combined}} = M_{\text{padding}} \wedge M_{\text{causal}}.$$

The combined mask has shape $(B, 1, T, T)$.

For the first sequence, with two real tokens:

$$M_1 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \end{bmatrix}.$$

For the second sequence, with four real tokens:

$$M_2 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix}.$$

Here, 1 means allowed, and 0 means blocked.

In the first sequence, positions 2 and 3 are padding, so they are blocked as keys. The causal mask also blocks future keys.

I checked all blocked positions in the attention weights. The largest blocked weight was:

$$\boxed{0.0}$$

This shows that padded and future keys received zero attention weight.

Next, I multiplied the padded value vectors by a large number and checked whether the outputs for real query tokens changed.

The maximum changes were:

$$\boxed{[0.0,\ 0.0]}$$

The outputs did not change. This shows that the padded values did not affect the results for real tokens.

I tested fully blocked rows separately in Section 2.1. They returned zeros without warnings or NaN values.

Mask rules in PyTorch: My NumPy code uses True for allowed keys.

In `torch.nn.MultiheadAttention`, True means blocked. Therefore, I invert my Boolean mask when using this PyTorch function.

The PyTorch key-padding mask also uses True for padding.

In `torch.nn.functional.scaled_dot_product_attention`, True means allowed, just like in my NumPy code.

For my PyTorch reference, I used `torch.nn.MultiheadAttention` and inverted the allowed mask.

I compared only rows with at least one allowed key because fully blocked rows need special handling.

Part 3: Multi-Head Attention

3.1 Shape trace, attention maps, and outputs

I used the provided token embeddings for The, cat, and sleeps:

$$X = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 0 \end{bmatrix}.$$

The input has one batch, three tokens, and four features:

$$B = 1, \quad T = 3, \quad D = 4.$$

I used two attention heads:

$$H = 2, \qquad d_h = \frac{D}{H} = 2.$$

I initialized the projection matrices using `initialize_mha(4, seed=577)`.

First, I projected the complete input using the query, key, and value matrices:

$$Q_{\text{full}} = XW_Q, \qquad K_{\text{full}} = XW_K, \qquad V_{\text{full}} = XW_V.$$

Next, I split the projected features into two heads and calculated attention separately for each head.

The shapes at each step were:

Stage	Meaning	Shape
X	Input features	$(1, 3, 4)$
XW_Q, XW_K, XW_V	Projected features	$(1, 3, 4)$
Q, K, V	Split heads	$(1, 2, 3, 2)$
S, A	Scores and weights	$(1, 2, 3, 3)$
Head outputs	Weighted values	$(1, 2, 3, 2)$
Merged outputs	Joined heads	$(1, 3, 4)$
Y_{MHA}	Final output	$(1, 3, 4)$

Each head calculates its own attention weights. The scaling factor is:

$$\sqrt{d_h} = \sqrt{2}.$$

The attention weights for Head 0 were:

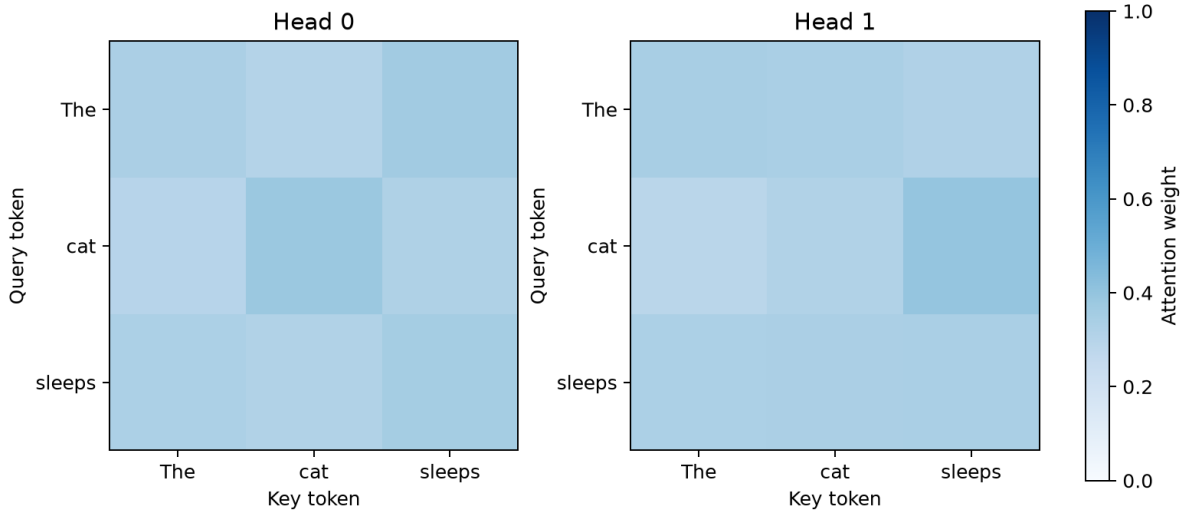
$$A_0 = \begin{bmatrix} 0.3336 & 0.3057 & 0.3607 \\ 0.2945 & 0.3826 & 0.3229 \\ 0.3310 & 0.3147 & 0.3544 \end{bmatrix}.$$

The attention weights for Head 1 were:

$$A_1 = \begin{bmatrix} 0.3433 & 0.3370 & 0.3198 \\ 0.2884 & 0.3139 & 0.3978 \\ 0.3293 & 0.3346 & 0.3361 \end{bmatrix}.$$

Both attention weight matrices have shape $(3, 3)$.

I plotted the weights for both heads. The rows show query tokens, and the columns show key tokens. Both plots use the same color scale from zero to one.



After calculating attention, I merged the two head outputs and applied the output projection:

$$Y_{\text{MHA}} = \text{Concat}(Y_0, Y_1)W_O.$$

The final output was:

$$Y_{\text{MHA}} = \begin{bmatrix} -0.8025 & 0.2970 & -0.2950 & 0.2870 \\ -0.7275 & 0.4051 & -0.2322 & 0.2978 \\ -0.7932 & 0.3091 & -0.2887 & 0.2858 \end{bmatrix}.$$

The final output has shape $(1, 3, 4)$. The matrix above shows the single batch.

Why projection comes before splitting: The projection matrices use all four input features to calculate new query, key, and value features.

If I split the original input before projection, each head would only receive part of the original features.

This would not be the same as using the learned full-width projections, because the projections can mix information from all input features. **Attention map interpretation:** The attention weights in both heads are mostly between 0.29 and 0.40. The heads give similar attention to all three words, without strongly focusing on one word. This makes sense because the projection matrices have random values and have not been trained to learn language patterns.

3.2 Independent PyTorch comparison

I compared my NumPy implementation with `torch.nn.MultiheadAttention`.

I used the following settings:

```
embed_dim = 4
num_heads = 2
bias = False
dropout = 0.0
batch_first = True
```

```
dtype = torch.float64
device = CPU
```

PyTorch stores linear weights in transposed form compared with my NumPy implementation.

I copied the query, key, and value matrices into PyTorch by stacking their transposes:

$$W_{\text{in}} = \begin{bmatrix} W_Q^\top \\ W_K^\top \\ W_V^\top \end{bmatrix}.$$

For the output projection, I used:

$$W_{\text{out}} = W_O^\top.$$

I used `average_attn_weights=False` to compare the weights for each head separately.

My NumPy code uses True for allowed positions. In PyTorch's `nn.MultiheadAttention`, True means blocked, so I inverted the mask.

I tested three cases: unmasked self-attention, causal self-attention, and rectangular cross-attention.

The cross-attention test used two query tokens and three context tokens.

The maximum absolute errors were:

Case	Output error	Weight error
Unmasked self	1.11×10^{-16}	5.55×10^{-17}
Causal self	1.11×10^{-16}	5.55×10^{-17}
Rectangular cross	1.11×10^{-16}	5.55×10^{-17}

All errors were smaller than 10^{-9}. This shows that my NumPy implementation agrees with the independent PyTorch reference within floating-point precision.

3.3 Head count and head-output ablation

I compared one attention head with two heads. I kept the input and all four projection matrices unchanged.

The results were:

Property	$H = 1$	$H = 2$
Features per head	4	2
Score shape	$(1, 1, 3, 3)$	$(1, 2, 3, 3)$
MHA parameters	64	64

The maximum output difference between one head and two heads was:

$$\boxed{0.0397}$$

Why the parameter count is the same: The total feature width stays $D = 4$. There are four projection matrices:

$$W_Q, W_K, W_V, W_O \in \mathbb{R}^{4\times 4}.$$

Each matrix has:

$$4 \times 4 = 16$$

parameters.

The total parameter count is:

$$4 \times 16 = \boxed{64}.$$

Changing the number of heads changes how the features are divided, but it does not change the size of these projection matrices.

Removing one head: I tested the effect of removing each head. I set one head's output to zero before merging the heads and applying W_O.

First, I checked that merging without removing a head gave the same output:

$$\text{Maximum difference} = 0.0.$$

Then I removed each head separately.

The maximum output changes were:

Experiment	Maximum output change
Remove Head 0	0.4549
Remove Head 1	0.4669

Both heads affect the final output. Removing either head changes the result.

Head 1 caused a slightly larger maximum change in this example. However, this does not prove that Head 1 is more important in general.

The projection matrices in this experiment are fixed sample parameters and were not trained on a language task.

Therefore, these attention maps do not prove that either head learned a specific meaning or language role.

Why the outputs can differ: With one head, all four features are used together to calculate attention weights. The scores are divided by $\sqrt{4} = 2$, and softmax is applied once.

With two heads, the features are split into two groups. Each head uses two features, divides the scores by $\sqrt{2}$, and calculates its own softmax.

Softmax is not a linear function, so two separate softmax calculations do not give the same result as one combined calculation.

That is why changing the number of heads can change the final output, even when the input and projection matrices stay the same.

Part 4: Transformer Block

4.1 Shape trace and reference checks

I used the provided function `initialize_block(4, 8, seed=577)` to create the parameters.

The Transformer block uses pre-normalization, two residual connections, and a feed-forward network with ReLU. No dropout is used.

The calculations are:

$$\begin{aligned} N_1 &= \mathrm{LN}_1(X), \\ U &= X + \mathrm{MHA}(N_1), \\ N_2 &= \mathrm{LN}_2(U), \\ F &= \mathrm{ReLU}(N_2 W_1 + b_1) W_2 + b_2, \\ Z &= U + F. \end{aligned}$$

The input has one batch, three tokens, and four features:

$$X : (1, 3, 4).$$

The shapes of the intermediate results are:

Variable	Shape
N_1	$(1, 3, 4)$
U	$(1, 3, 4)$
N_2	$(1, 3, 4)$
F	$(1, 3, 4)$
Z	$(1, 3, 4)$
Attention weights	$(1, 2, 3, 3)$

The feed-forward network has a hidden width of 8. The parameter shapes are:

$$\begin{aligned} W_1 &: (4, 8), \quad b_1 : (8), \\ W_2 &: (8, 4), \quad b_2 : (4). \end{aligned}$$

Each LayerNorm uses scale and bias vectors of shape (4). It uses population variance and $\epsilon = 10^{-5}$.

The final Transformer block output was:

$$Z = \begin{bmatrix} 1.6988 & -1.1299 & 0.4089 & 0.9216 \\ 0.9056 & 0.7420 & 1.5733 & -1.1551 \\ 0.6542 & 0.9556 & -0.1768 & -0.0796 \end{bmatrix}.$$

The final output has shape $(1, 3, 4)$. The matrix above shows the single batch.

Independent PyTorch reference check: I created the same Transformer block using PyTorch LayerNorm, my independent PyTorch multi-head attention reference, and PyTorch matrix multiplication with ReLU.

I used the same input and parameters in both implementations.

The maximum absolute output difference was:

$$\boxed{6.66 \times 10^{-16}}$$

This is smaller than the required 10^{-9}, so my NumPy and PyTorch results agree within floating-point precision.

Zero-update residual check: I set the following parameters to zero:

$$W_O = 0, \qquad W_2 = 0, \qquad b_2 = 0.$$

Setting W_O to zero makes the attention output zero, so the first residual connection returns the original input:

$$U = X + 0 = X.$$

Setting W_2 and b_2 to zero makes the feed-forward output zero:

$$F = 0.$$

Therefore, the second residual connection gives:

$$Z = U + 0 = X.$$

The maximum absolute difference between the block output and the original input was:

$$\boxed{0.0}$$

This confirms that the residual connections allow the block to return the input unchanged when both updates are zero.

4.2 Positions and reordering

I used sinusoidal position vectors to give the model information about token positions.

For an even feature width D, the formulas are:

$$\omega_m = 10000^{-2m/D},$$

$$PE_{p,2m} = \sin(p\omega_m), \qquad PE_{p,2m+1} = \cos(p\omega_m).$$

For $D = 4$, the two rates are:

$$\omega_0 = 1, \qquad \omega_1 = 0.01.$$

I created position vectors for three positions. The position matrix has shape $(3, 4)$:

$$PE = \begin{bmatrix} 0 & 1 & 0 & 1 \\ 0.8415 & 0.5403 & 0.0100 & 1.0000 \\ 0.9093 & -0.4161 & 0.0200 & 0.9998 \end{bmatrix}.$$

Each row represents one position, and each position has four features.

I added the position vectors to the token embeddings feature by feature:

$$X_{\text{pos}} = X + PE.$$

The input X has shape $(1, 3, 4)$, and PE has shape $(3, 4)$. NumPy broadcasts PE over the batch dimension.

The result X_{pos} has shape $(1, 3, 4)$. The position vectors are added, not concatenated, so the number of features stays the same.

Reordering without position vectors: I changed the token order from The/cat/sleeps to sleeps/The/cat, using indices $[2, 0, 1]$.

I tested the block without position vectors and without a causal mask.

The maximum absolute difference was:

$$\boxed{5.55 \times 10^{-16}}$$

This is almost zero. Attention uses information from all tokens and does not depend on their order when no position information is given.

LayerNorm and the feed-forward network work on each token separately.

Because of this, changing the input order changes the output order in the same way. This is called permutation equivariance.

It means the block cannot identify token order without position information.

Reordering with position vectors: Next, I added position vectors and repeated the same test.

I kept the position vectors in their original slots and moved only the token embeddings.

The maximum absolute difference was:

$$\boxed{1.2616}$$

Each moved token now receives a different position vector. Its new input is the token embedding plus the position vector at its new location.

Therefore, the new input is not just a reordered copy of the original input.

This changes the attention calculations and the final outputs.

The result shows that position vectors make the Transformer block sensitive to token order.

4.3 Future perturbation and causal visibility

I used the input with position vectors:

$$X_{\text{pos}} = X + PE.$$

I changed only the last token by adding the following vector to its embedding:

$$\Delta = [5, -3, 2, -4].$$

I compared the outputs at the first two token positions before and after this change.

With a causal mask: The maximum absolute output difference at the first two positions was:

$$\boxed{0.0}$$

With a causal mask, the first two tokens cannot attend to the last token because it is in the future.

LayerNorm and the feed-forward network work on each token separately. The residual connections also keep the information at the same token position.

Therefore, changing the last token does not affect the first two outputs.

Without a causal mask: The maximum absolute output difference at the first two positions was:

$$\boxed{0.5173}$$

Without a causal mask, the first two tokens can attend to the last token.

Changing the last token can change its key and value vectors. This can affect the attention weights and the outputs of the earlier tokens.

LayerNorm, the feed-forward network, and residual connections do not directly mix information between different tokens. The change reaches earlier positions through self-attention.

Why the diagonal is allowed: When the model uses position i to predict token $i + 1$, it is allowed to see the current token at position i.

The diagonal of the causal mask allows a token to attend to itself.

It does not allow the token to see future positions, including position $i + 1$.

Therefore, allowing the diagonal does not reveal the token being predicted.

Part 5: Architecture Comparison

5.1 Encoder–decoder and decoder-only generation

We want to translate the English sentence "The cat sleeps" into French: "Le chat dort".

First, the encoder reads all three source tokens: The, cat, and sleeps. It creates a representation for each source token.

The decoder starts with the supplied BOS (beginning-of-sequence) token.

To predict the first word, Le, the decoder uses the BOS position.

In cross-attention, the query comes from the decoder's current hidden representation at the BOS position.

The keys and values come from the encoder representations of all three source tokens: The, cat, and sleeps.

The decoder uses these source representations to help predict Le.

After Le is predicted, it becomes the next decoder input.

To predict chat, the decoder uses the representation at the Le position.

With causal self-attention, the decoder can see the target inputs BOS and Le, but it cannot see future target words such as chat or dort.

In cross-attention, it can still use all three source tokens: The, cat, and sleeps.

In this example, BOS is supplied to start generation. It is not predicted by the model.

Decoder-only Transformer: For the decoder-only model, the prompt is:

$$\text{French : The cat sleeps =}$$

All prompt tokens are given to the model.

The final hidden vector at the "=" position is used to predict the next token, Le.

During causal prefill, the model processes the prompt tokens together, but each query can only attend to its own position and earlier positions.

For example, the query at the The position cannot attend to the later cat or sleeps positions.

This is true even though the full prompt is already available to the program.

The causal mask controls which keys each query can use. Having all tokens in memory does not mean every query can read all of them.

Main difference: The encoder-decoder model uses an encoder to read the full source sentence and a decoder to generate the target sentence.

The decoder uses cross-attention to get information from the source sentence.

The decoder-only model uses one causal sequence containing the prompt and the generated tokens.

It predicts the next token using the allowed earlier positions in that sequence.

Assistance and Generative-AI Disclosure

```
python -m unittest -v test_hw3_public.py
```

What I did and verified myself: I wrote the core implementation of all eight functions and the notebook TODOs, performed the hand calculations and the variance derivation, filled in all numerical values, ran the public tests and all notebook checks, and wrote the explanations in this report. I verified every reported number against my own notebook output.

I used ChatGPT: to help me with English grammar and LaTeX formatting.